

Chapter 11 - Coordinates in Space

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#Space

A space is the set of all possible points that have three coordinates. Alternatively, A space can be thought as the combination of infinite planes stacked on each other parallelly. It is studied by considering three mutually perpendicular lines: x-axis, y-axis and z-axis which intersect at a fixed point called origin.

Right-hand-thumb rule

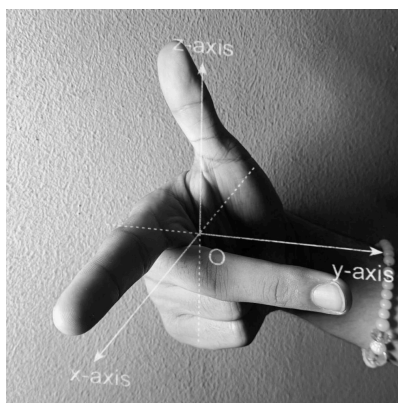
Here,

Index finger gives the direction of x-axis

Middle finger gives the direction of y-axis

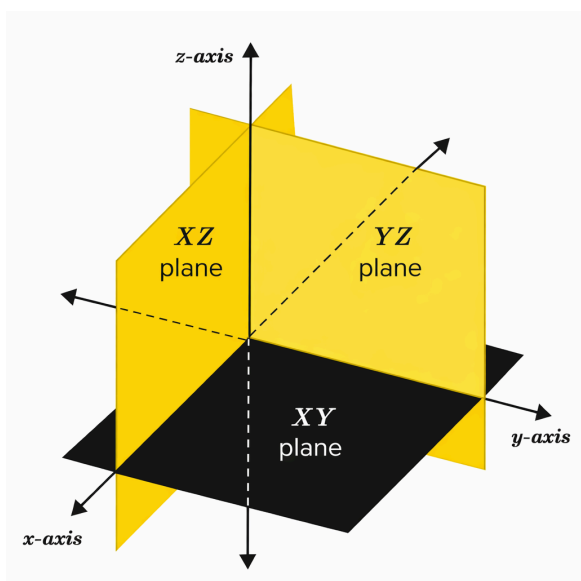
Thumb gives the direction of z-axis

This is known as the right-hand-thumb rule.

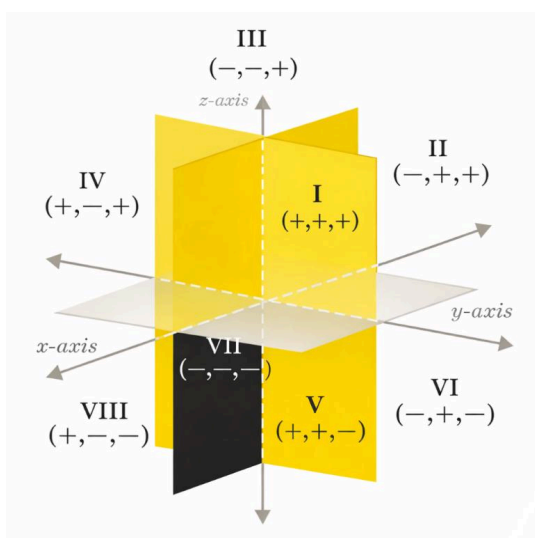


Octants

Octants are the eight regions formed by the intersection of three axial planes XY, YZ and XZ.



The signs of the coordinates of any point can be found by applying the right-hand-thumb rule.

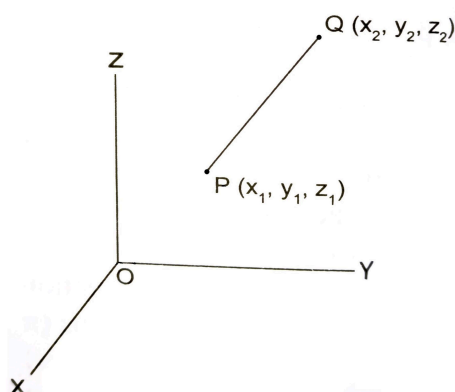


$\times$	Octant 1	Octant 2	Octant 3	Octant 4	Octant 5	Octant 6	Octant 7	Octant 8
x	+	-	-	+	+	-	-	+
y	+	+	-	-	+	+	-	-
z	+	+	+	+	-	-	-	-

Ordered Triple

An ordered triple is an ordered list of 3 real numbers in form of (x, y, z) . It is used to represent the coordinates of any point in the space.

#Distance between Two Points



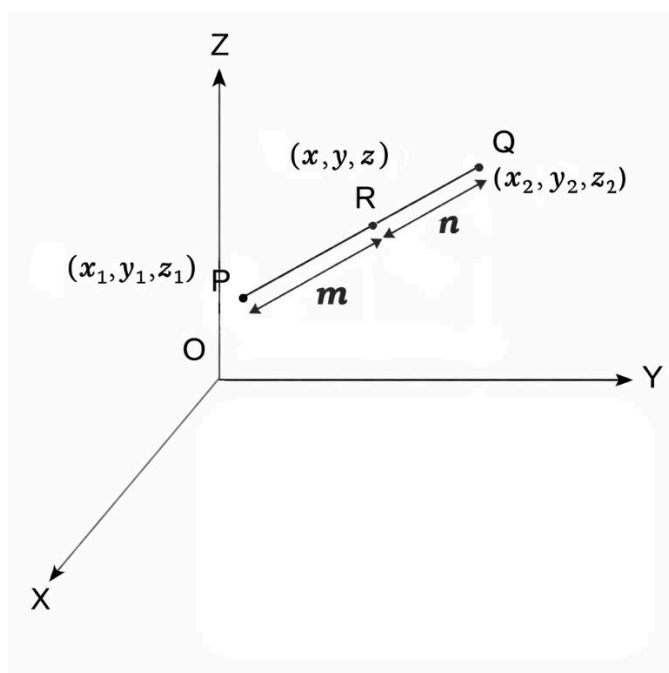
The distance between two points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ is

$$PQ = d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Corollary: The distance between the origin and the point (x, y, z) is $d = \sqrt{x^2 + y^2 + z^2}$

#Section Formula

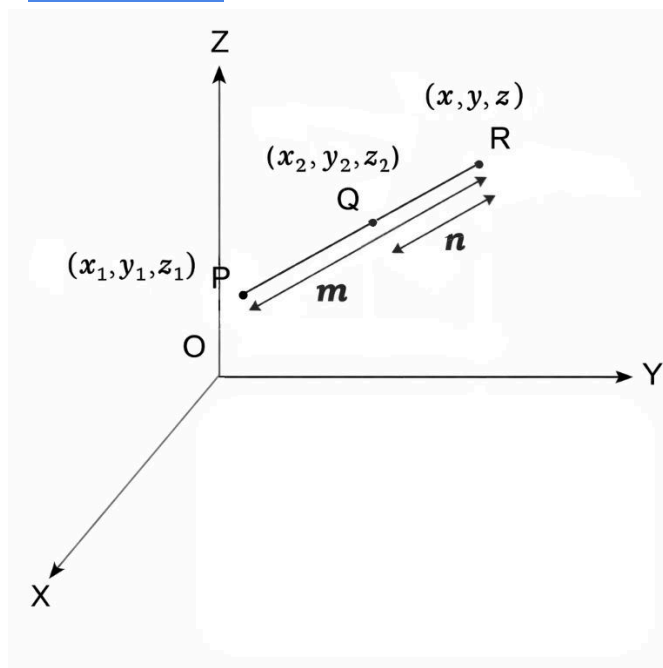
Internal Division



If two points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ are internally divided by a point $R(x, y, z)$ in the ratio $m : n$ then

$$(x, y, z) = \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}, \frac{mz_2 + nz_1}{m+n} \right)$$

External Division

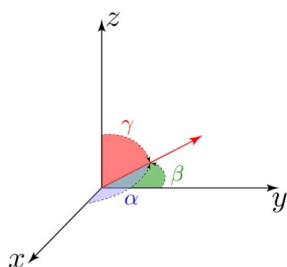


If two points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ are external divided by a point $R(x, y, z)$ in the ratio $m : n$ then

$$(x, y, z) = \left(\frac{mx_2 - nx_1}{m-n}, \frac{my_2 - ny_1}{m-n}, \frac{mz_2 - nz_1}{m-n} \right)$$

#Direction Cosines of a Line

The cosine of angles made by any line with X, Y and Z axis are known as direction cosines (d.c.s.).



If a line makes angles α , β and γ with x-axis, y-axis and z-axis respectively then $\cos \alpha$, $\cos \beta$ and $\cos \gamma$ are called the direction cosines of the given line.

$$l = \cos \alpha \quad m = \cos \beta \quad n = \cos \gamma$$

Relation between the direction cosines of a line:

$$l^2 + m^2 + n^2 = 1$$

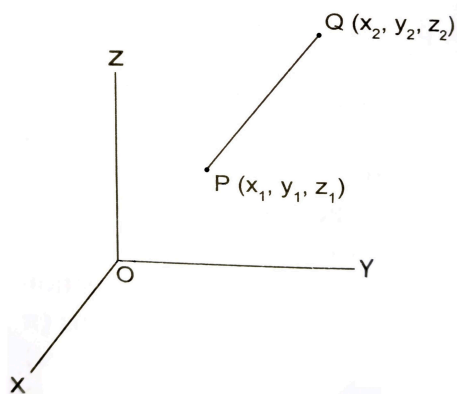
Direction cosines of the coordinate axes

d.c.s. of x-axis are 1, 0, 0

d.c.s. of y-axis are 0, 1, 0

d.c.s. of z-axis are 0, 0, 1

Direction cosines of the Line through Two Given Points

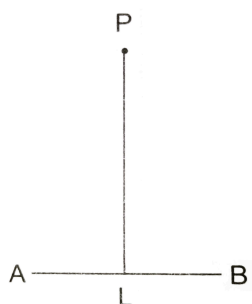


$$\frac{x_2 - x_1}{r}, \frac{y_2 - y_1}{r}, \frac{z_2 - z_1}{r} \text{ where } r = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

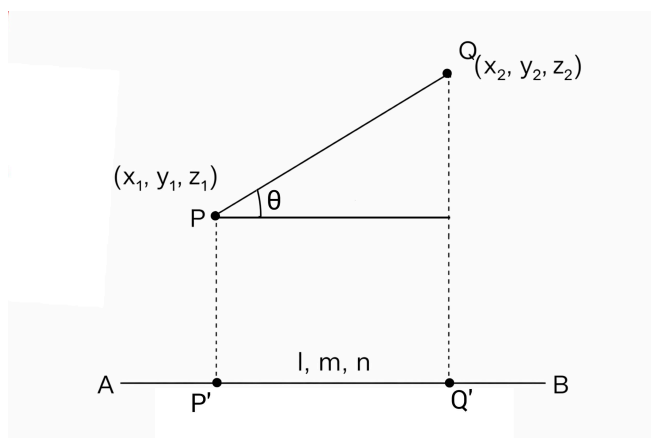
$$\text{Also, } \frac{x_2 - x_1}{\cos \alpha} = \frac{y_2 - y_1}{\cos \beta} = \frac{z_2 - z_1}{\cos \gamma} = r$$

Projection

Projection of a point on a line or a plane



Projection of a line joining two points on the another line



Projection of PQ on AB = $P'Q' = (x_2 - x_1)l + (y_2 - y_1)m + (z_2 - z_1)n$

Also, $P'Q' = PQ \cos \theta$

Direction ratios

If a set of numbers a, b, c are proportional to the d.c.s. l, m, n then they are called direction ratios (d.r.s) of the

given line. i.e., $\frac{l}{a} = \frac{m}{b} = \frac{n}{c}$

Relation between the direction ratios and direction cosines

If a, b, c are the direction ratios and l, m, n are the direction cosines of a line then

$$l = \frac{a}{\sqrt{a^2 + b^2 + c^2}} \quad m = \frac{b}{\sqrt{a^2 + b^2 + c^2}} \quad n = \frac{c}{\sqrt{a^2 + b^2 + c^2}}$$

Direction ratios of the Line through Two Given Points

The direction ratios of the line PQ through two given points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ are

$$x_2 - x_1, y_2 - y_1, z_2 - z_1$$

Angle between two lines

$$\cos \theta = l_1 l_2 + m_1 m_2 + n_1 n_2$$

Expression for $\sin \theta$

$$\sin \theta = \pm \sqrt{(l_1 m_2 - l_2 m_1)^2 + (m_1 n_2 - m_2 n_1)^2 + (n_1 l_2 - n_2 l_1)^2}$$

Condition of perpendicularity of two lines

$$l_1 l_2 + m_1 m_2 + n_1 n_2 = 0$$

Condition of parallelism of two lines

$$l_1 = l_2, m_1 = m_2 \text{ and } n_1 = n_2$$

Angle between two straight lines whose direction ratios are given

$$\cos \theta = \frac{a_1 a_2 + b_1 b_2 + c_1 c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}}$$

1. Condition of perpendicularity: $a_1 a_2 + b_1 b_2 + c_1 c_2 = 0$

2. Condition of parallelism: $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$